

SE LAB-1

PROBLEM STATEMENT #48

Incident Escalation & On-Call Rotation Engine

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REQUIREMENTS

Functional Requirements

FR-001

Description-the software should be able to monitor alerts and be able identify active on-call engineer from the rotation schedule.

Priority-High

Acceptance Criteria

Pass-Alert should be assigned to active on-call engineer.

Fail-Alert gets assigned to unavailable engineer.

Rationale-This ensures that alerts are immediately routed to the correct active engineer.

FR-002

Description- Any incident which has not been acknowledged should be escalated to any secondary engineer after 5 minutes.

Priority-High

Acceptance Criteria-

Pass-Incidents that are Unacknowledged are automatically escalated after 5 minutes

Fail-Incident remains with primary engineer without escalation

Rationale- Prevents any critical incident from remaining unattended.

FR-003

Description- When an incident is assigned to an engineer they should be notified through Phone/SMS escalation ladders.

Priority-High

Acceptance Criteria-

Pass-Engineer is notified successfully through the channel about the incident.

Fail-The System fails to send notification through the desired channel.

Rationale-Speedy communication medium through which engineers can respond quickly.

FR-004

Description-Incident commander should be able to track incident status and acknowledgements.

Priority-Medium

Acceptance Criteria

Pass-the Incident commander is able to view all details related to the incident.

Fail-Incident related information is unavailable.

Rationale-Enables effective monitoring during the incidents.

FR-005

Description -The system should display all incident related information like post-mortem information, including incident details, actions taken, escalation events, and resolution.

Priority-Medium

Acceptance Criteria-

Pass- A complete post-mortem can be recorded and retrieved

Fail- Required incident or resolution information cannot be stored or retrieved.

Rationale-Helps in incident analysis and also helps improve future incident response

Non Functional Requirements

NFR-001-Performance & Security

Description- Alert dispatch through webhook, SMS, and email shall be initiated within 3 seconds of alert ingestion under expected load.

Priority-High

Acceptance Criteria

Pass: Benchmark testing shows alert dispatch initiation within 3 seconds.

Fail: Dispatch exceeds the 3-second target under simulated peak load.

Rationale- Rapid alert delivery is essential for timely incident response.

NFR-002- Reliability & Availability

Description- The system shall maintain reliable alert routing and escalation without losing incidents or escalation events during normal and simulated peak operation.

Priority-High

Acceptance Criteria

Pass: All test alerts and escalation events are successfully recorded and processed.

Fail: Any alert or escalation event is lost or incorrectly processed.

Rationale- Prevents missed incidents and ensures dependable operation of the on-call system.

